Marks: 2 (Budgeted Time 4 Min)

How can the dangling reference problem be avoided ?

Answer (Please click here to Add Answer)

Normal Arial 12 100%

To avoid dangling reference, don’t return the reference of a local variable (transient) from a function.

Question No : 22 of 26

Min)

What is "level" in binary tree ?

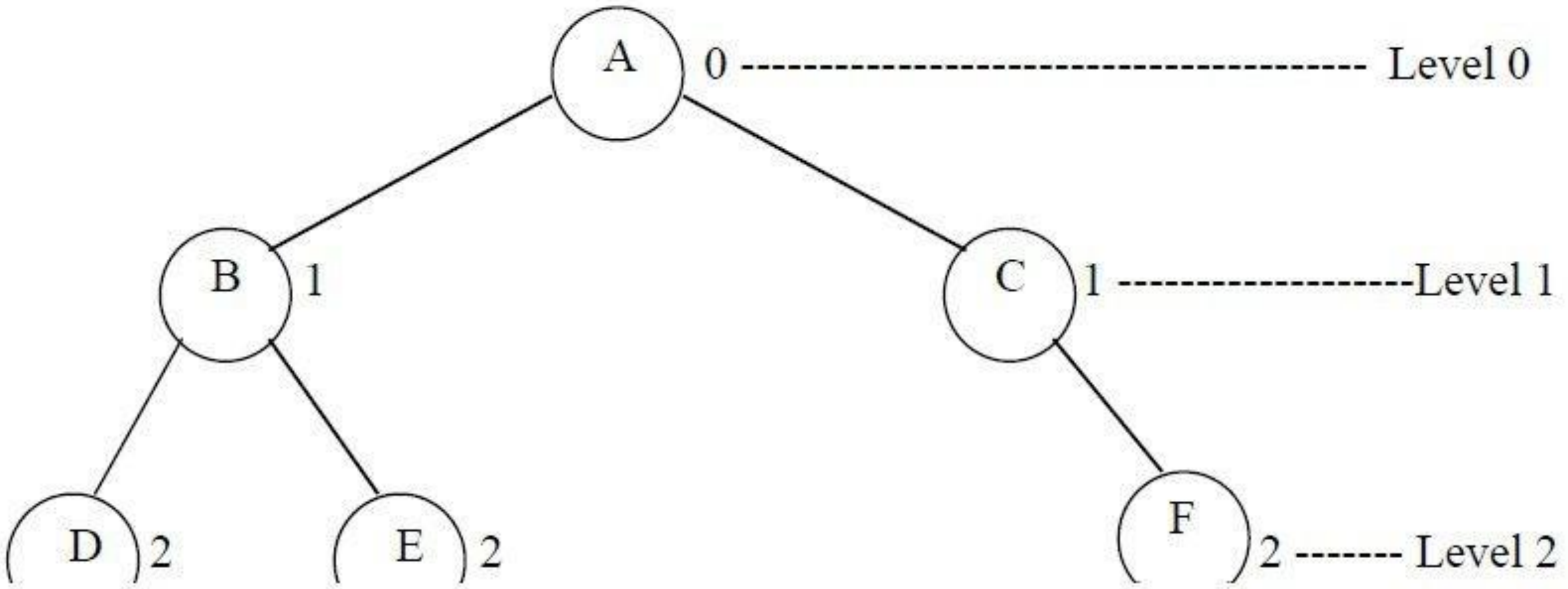
Answer (Please click here to Add Answer

Normal Arial 12

The level of a node in a binary tree is defined as follows:

- Root has level 0,
- Level of any other node is one more than the level its parent (father).
- The *depth* of a binary tree is the maximum level of any leaf in the tree.

To understand level of a node, consider the following figure 11.13. This figure shows the same tree of figure 11.2, discussed so far.



Question No : 23 of 26 … 3 (Budgeted Time 6 Min)

Draw a BST that is as tall as possible and contains the values '2', '6', '5', '4', '7', '9', '8'.

Answer (Please click here to Add Answer)

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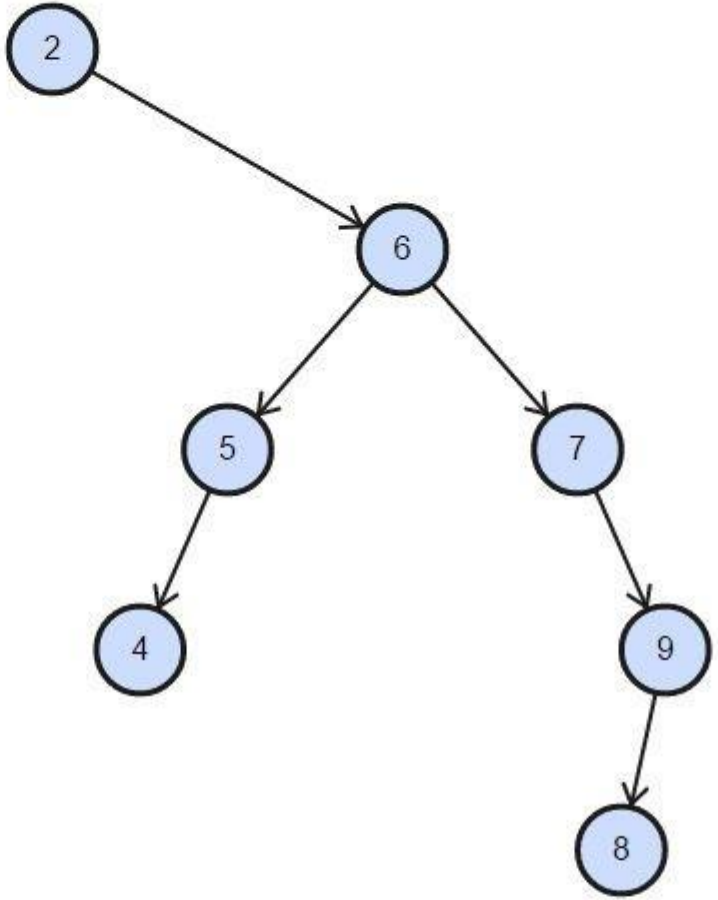

Question No : 24 of 26 Marks: 3 (Budgeted Time 6 Min)

Define the following terms

1) Reference variables
2) Dangling reference
3) Const keyword

Answer (Please click here to Add Answer)

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Ref Vari: A reference variable is an alias, that is, another name for an already existing variable. Once a reference is initialized with a variable, either the variable name or the reference name may be used to refer to the variable.
Dangling: A dangling reference is a reference to an object that no longer exists.
Const: By writing *const*, we are saying that parameter must remain constant for the life of the function.

Question No : 25 of 26 Marks: 5 (Budgeted Time 10 Min)

Show the result of inserting the values **35, 19, 41, 39, 37, 28, 31, 14** into an empty AVL Tree.
You have to show only the complete AVL tree, steps are not required.

Answer (Please click here to Add Answer)

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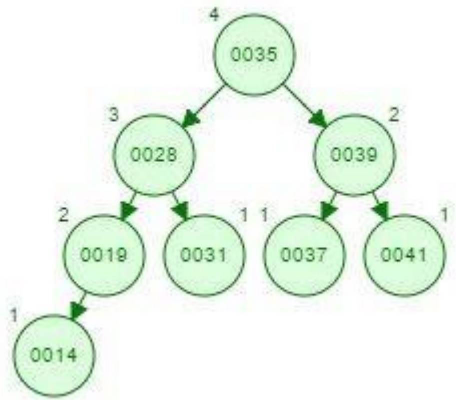

Question No : 26 of 26 Marks: 5 (Budgeted Time 10 Min)

Write down the algorithm (Steps) for level order traversing of a binary tree.

Answer (Please cli

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The name of the method is *levelorder* and it is accepting a pointer of type *TreeNode <int>*. This method will start traversing tree from the node being pointed to by this pointer. The first line (line 3) in this method is creating a queue *q* by using the *Queue* class factory interface. This queue will be containing the *TreeNode<int>* * type of objects. Which means the queue will be containing nodes of the tree and within each node the element is of type *int*.

The line *5* is checking to see, if the *treeNode* pointer passed to the function is pointing to NULL. In case it is NULL, there is no node to traverse and the method will return immediately.

Otherwise at line *6*, the very first node (the node pointed to by the *treeNode* pointer) is added in the queue *q*.

Next is the *while* loop (at line *7*), which runs until the queue *q* does not become empty. As we have recently added one element (at line *6*), so this loop is entered.

At line *9*, *dequeue()* method is called to remove one node from the queue, the element at *front* is taken out. The return value will be a pointer to *TreeNode*. In the next line (line *10*), the *int* value inside this node is taken out and printed. At line *11*, we check to see if the left subtree of the tree node (we've taken out in at line *9*) is present. In case the left subtree exists, it is inserted into the queue in the next statement at line *12*. Next, we see if the right subtree of the node is there, it is inserted into the queue also. Next statement at line *15* closes the *while* loop. The control goes back to the line *7*, where it checks to see if there is some element left in the queue. If it is not empty, the loop is entered again until it becomes empty.

Question No : 21 of 26 Marks: 2 (Budgeted Time 4 Min)

In which of the traversal methods, recursion can not be applied?

Answer (Please click here to Add Answer)

Normal Arial 12 100%

by using an explicit stack.
We cannot avoid stack. The stack will be used to store the tree nodes in the appropriate order.

Question No : 22 of 26 Marks: 2 (Budgeted Time 4 Min)

How can we calculate the height of tree ?

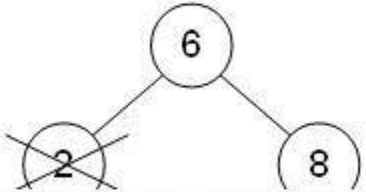
Answer (Please click here to Add Answer)

Normal Arial 12 100%

It is the longest path from the node to a leaf. So height is the number of edges of the path

Question No : 23 of 26 Marks: 3 (Budgeted Time 6 Min)

What will be the resultant tree after deleting the node 2 from the tree given below.



Answer (Please click here to Add Answer)

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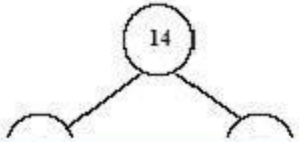
Question No : 24 of 26 Marks: 3 (Budgeted Time 6 Min)

Consider the following AVL tree. Insert a new node with key of **12**. No need to show all steps, just draw the final AVL tree .



Answer (Please click here to Add Answer)

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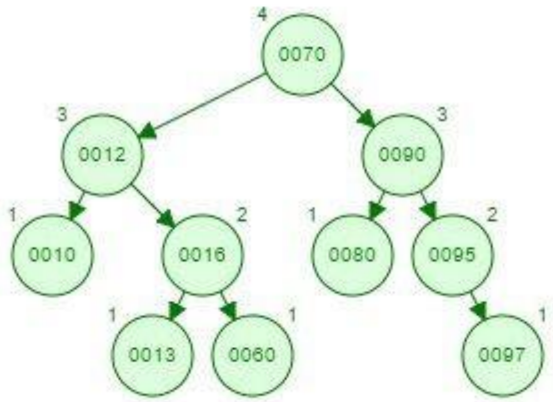
See AVL tree
youtube

Question No : 25 of 26 Marks: 5 (Budgeted Time 10 Min)

Show the result of inserting the values **10, 80, 12, 70, 90, 60, 13, 16, 95, 97** into an empty AVL Tree. You have to show only the complete AVL tree, steps are not required.

Answer (Please



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Question No : 26 of 26 Marks: 5 (Budgeted Time 10 Min)

Explain the priority queue with the help of real world examples?

Answer (Please click here to Add Answer)

the queue is a FIFO (First in first out) structure. In daily life, you have also seen that it is not true that a person, who comes first, leaves first from the queue. Let's take the example of traffic. Traffic is stopped at the signal. The vehicles are in a queue. When the signal turns green, vehicles starts moving. The vehicles which are at the front of the queue will cross the crossing first. Suppose an ambulance comes from behind. Here ambulance should be given priority. It will bypass the queue and cross the intersection. Sometimes, we have queues that are not FIFO i.e. the person who comes first may not leave first. We can develop such queues in which the condition for leaving the queue is not to enter first. There may be some priority. Here we will also see the events of future like the customer is coming at what time and leaving at what time. We will arrange all these events and insert them in a priority queue. We will develop the queue in such a way that we will get the event which is going to happen first of all in the future. This data structure is known as priority queue. In a sense, FIFO is a special case of priority queue in which priority is given to the time of arrival. That means the person who comes first has the higher priority while the one who comes later, has the low priority. You will see the priority queue being used at many places especially in the operating systems. In operating systems, we have queue of different processes. If some process comes with higher priority will be processed first

Question No : 21 of 26 Marks: 2 (Budgeted Time 4 Min)

How can we calculate the height of tree ?

Answer (Please click here to Add Answer)

Normal Arial 12 100%

It is the longest path from the node to a leaf. So height is the number of edges of the path

Question No : 22 of 26

Marks: 2 (Budgeted Time 4 Min)

What is meant by traversing of binary tree?

traverse

/ˈtravəs,trəˈvəːs/

verb

gerund or present participle: **traversing**

1. travel across or through.
   "he traversed the forest"
   synonyms: travel over/across, cross, journey over/across, across, negotiate; More
2. move back and forth or sideways.
   "a probe is traversed along the tunnel"

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"A binary tree is a finite set of elements that is either empty or is partitioned into three disjoint subsets. The first subset contains a single element called the root of the tree.

Question No : 23 of 26 Marks: 3 (Budgeted Time 6 Min)

Consider the following code snippet,

```
class BST_node
 {
   private:
    char ch;
```

Answer (Please click here to Add Answer)

Normal Arial 12 100%

Question No : 24 of 26 Marks: 3 (Budgeted Time 6 Min)

Explain the logic of the following function.

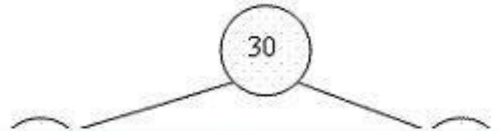
```
int&  Fun(int&  y)
   {
     y = y +100;
 return y;
```

Answer (Please click here to Add Answer)

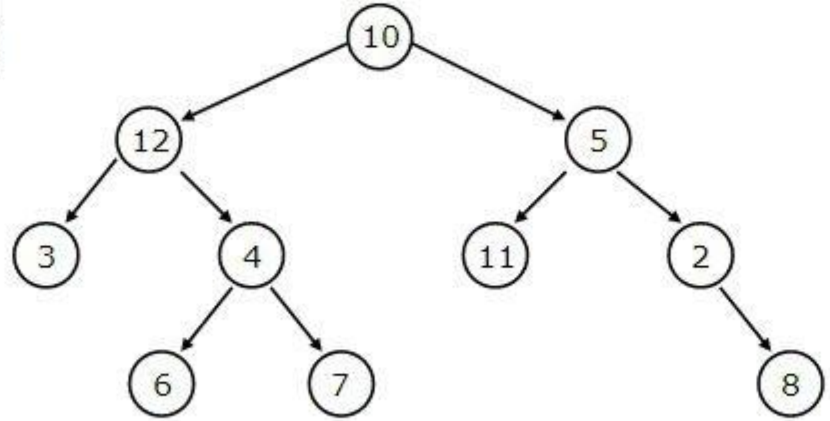
Normal Arial 12 100%

Question No : 25 of 26 Marks: 5 (Budgeted Time 10 Min)

Perform the Preorder and Inorder traversal on the tree given below and show the result in both cases (you have to perform the traversal manually not using any programming language)



Answer (Please click here to Add Answer)



Levelorder tree traversal
10, 12, 5, 3, 4, 11, 2, 6, 7, 8

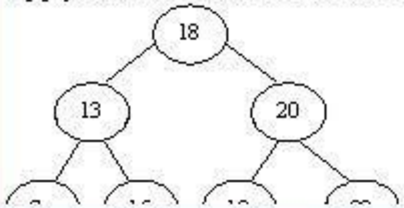
Inorder tree traversal
3, 12, 6, 4, 7, 10, 11, 5, 2, 8

Preorder tree traversal
10, 12, 3, 4, 6, 7, 5, 11, 2, 8

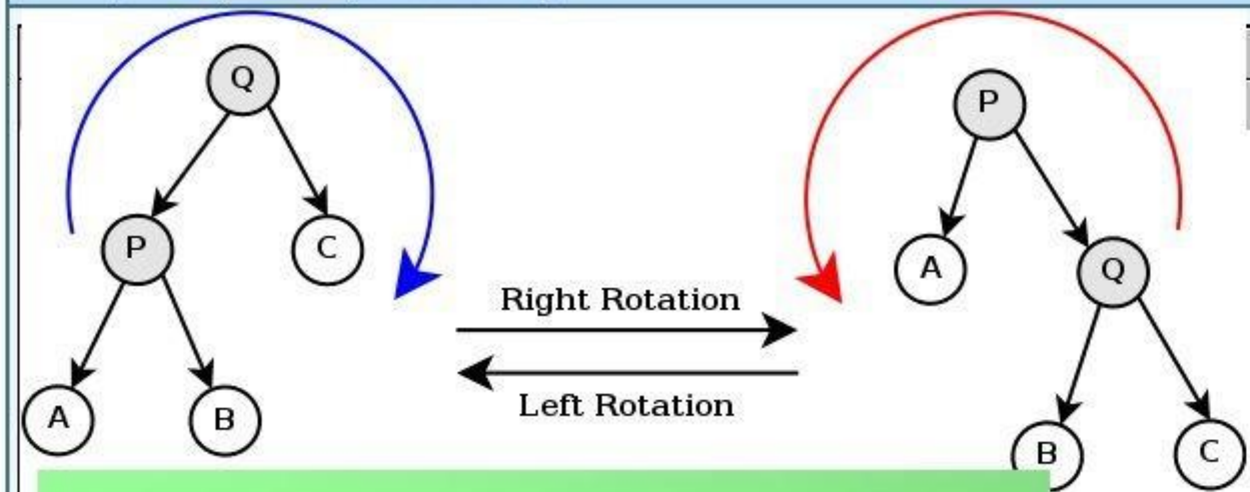
Postorder tree traversal
3, 6, 7, 4, 12, 11, 8, 2, 5, 10

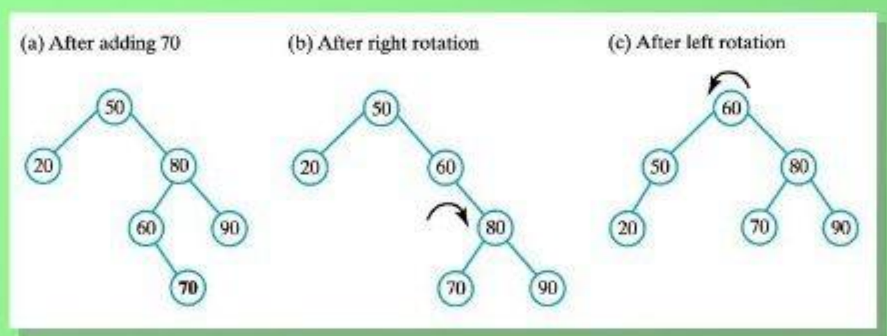
Question No : 26 of 26 Marks: 5 (Budgeted Time 10 Min)

Apply double rotation on the following tree to restore its balance. Show the necessary steps to perform this rotation.



Answer (Please click here to Add Answer)





(a) Adding 70 to the previous tree destroys its balance; to restore the balance, perform both (b) a right rotation and (c) a left rotation.

 Marks: 3 (Budgeted Time 6 Min)

```
    char ch;
    BST_node *BST_letf;
    BST_node *BST_right;
   };
 BST_node  *bst, bt;
```

Answer (Please click here to Add Answer)

Normal Arial 12 100%

Question No : 23 of 26 Marks: 3 (Budgeted Time 6 Min)

Correct the following statements,

```
bst.BST_left = NULL;
bst.BST_rifht = NULL;
bt->ch = 'A';
```

Answer (Please click here to Add Answer)

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